Program 7

Author

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Develop a function in R to plot a function curve based on a mathematical equation provided as input, with different curve style for each group, using ggplot2.

Step 1: Load Required Library

```
library(ggplot2)
```

Step 2: Create Data for the Function

```
# create a sequence of x values ranging from -2I to 2I
x <- seq(-2*pi, 2*pi, length.out = 500)

y1 <- sin(x)
y2 <- cos(x)

df <- data.frame(
  x = rep(x,2),
  y = c(y1, y2),
  group = rep(c("sin(x)", "cos(x)"), each = length(x))
)
df
```

```
                x             y  group
1    -6.28318531  2.449213e-16 sin(x)
2    -6.25800220  2.518045e-02 sin(x)
3    -6.23281909  5.034492e-02 sin(x)
4    -6.20763598  7.547747e-02 sin(x)
5    -6.18245288  1.005622e-01 sin(x)
6    -6.15726977  1.255831e-01 sin(x)
7    -6.13208666  1.505244e-01 sin(x)
8    -6.10690356  1.753702e-01 sin(x)
9    -6.08172045  2.001048e-01 sin(x)
10   -6.05653734  2.247125e-01 sin(x)
11   -6.03135423  2.491777e-01 sin(x)
12   -6.00617113  2.734849e-01 sin(x)
13   -5.98098802  2.976186e-01 sin(x)
14   -5.95580491  3.215637e-01 sin(x)
15   -5.93062180  3.453048e-01 sin(x)
16   -5.90543870  3.688269e-01 sin(x)
17   -5.88025559  3.921151e-01 sin(x)
18   -5.85507248  4.151547e-01 sin(x)
19   -5.82988937  4.379310e-01 sin(x)
20   -5.80470627  4.604296e-01 sin(x)
21   -5.77952316  4.826362e-01 sin(x)
22   -5.75434005  5.045367e-01 sin(x)
23   -5.72915694  5.261173e-01 sin(x)
24   -5.70397384  5.473642e-01 sin(x)
25   -5.67879073  5.682640e-01 sin(x)
26   -5.65360762  5.888035e-01 sin(x)
27   -5.62842451  6.089695e-01 sin(x)
28   -5.60324141  6.287494e-01 sin(x)
29   -5.57805830  6.481306e-01 sin(x)
30   -5.55287519  6.671007e-01 sin(x)
31   -5.52769208  6.856478e-01 sin(x)
32   -5.50250898  7.037601e-01 sin(x)
```

33	-5.47732587	7.214261e-01	sin(x)
34	-5.45214276	7.386346e-01	sin(x)
35	-5.42695965	7.553746e-01	sin(x)
36	-5.40177655	7.716357e-01	sin(x)
37	-5.37659344	7.874074e-01	sin(x)
38	-5.35141033	8.026798e-01	sin(x)
39	-5.32622722	8.174432e-01	sin(x)
40	-5.30104412	8.316882e-01	sin(x)
41	-5.27586101	8.454057e-01	sin(x)
42	-5.25067790	8.585872e-01	sin(x)
43	-5.22549479	8.712241e-01	sin(x)
44	-5.20031169	8.833086e-01	sin(x)
45	-5.17512858	8.948329e-01	sin(x)
46	-5.14994547	9.057897e-01	sin(x)
47	-5.12476236	9.161722e-01	sin(x)
48	-5.09957926	9.259736e-01	sin(x)
49	-5.07439615	9.351879e-01	sin(x)
50	-5.04921304	9.438090e-01	sin(x)
51	-5.02402993	9.518317e-01	sin(x)
52	-4.99884683	9.592507e-01	sin(x)
53	-4.97366372	9.660615e-01	sin(x)
54	-4.94848061	9.722596e-01	sin(x)
55	-4.92329751	9.778411e-01	sin(x)
56	-4.89811440	9.828026e-01	sin(x)
57	-4.87293129	9.871407e-01	sin(x)
58	-4.84774818	9.908529e-01	sin(x)
59	-4.82256508	9.939368e-01	sin(x)
60	-4.79738197	9.963903e-01	sin(x)
61	-4.77219886	9.982119e-01	sin(x)
62	-4.74701575	9.994006e-01	sin(x)
63	-4.72183265	9.999554e-01	sin(x)
64	-4.69664954	9.998761e-01	sin(x)
65	-4.67146643	9.991628e-01	sin(x)
66	-4.64628332	9.978158e-01	sin(x)
67	-4.62110022	9.958361e-01	sin(x)
68	-4.59591711	9.932248e-01	sin(x)
69	-4.57073400	9.899837e-01	sin(x)
70	-4.54555089	9.861148e-01	sin(x)
71	-4.52036779	9.816205e-01	sin(x)
72	-4.49518468	9.765037e-01	sin(x)
73	-4.47000157	9.707677e-01	sin(x)
74	-4.44481846	9.644161e-01	sin(x)
75	-4.41963536	9.574528e-01	sin(x)
76	-4.39445225	9.498824e-01	sin(x)
77	-4.36926914	9.417097e-01	sin(x)
78	-4.34408603	9.329397e-01	sin(x)
79	-4.31890293	9.235781e-01	sin(x)
80	-4.29371982	9.136308e-01	sin(x)
81	-4.26853671	9.031041e-01	sin(x)
82	-4.24335360	8.920047e-01	sin(x)
83	-4.21817050	8.803397e-01	sin(x)
84	-4.19298739	8.681164e-01	sin(x)
85	-4.16780428	8.553425e-01	sin(x)
86	-4.14262117	8.420263e-01	sin(x)
87	-4.11743807	8.281760e-01	sin(x)
88	-4.09225496	8.138006e-01	sin(x)
89	-4.06707185	7.989091e-01	sin(x)
90	-4.04188874	7.835109e-01	sin(x)
91	-4.01670564	7.676159e-01	sin(x)
92	-3.99152253	7.512341e-01	sin(x)
93	-3.96633942	7.343759e-01	sin(x)
94	-3.94115631	7.170520e-01	sin(x)
95	-3.91597321	6.992734e-01	sin(x)
96	-3.89079010	6.810513e-01	sin(x)
97	-3.86560699	6.623974e-01	sin(x)
98	-3.84042389	6.433233e-01	sin(x)
99	-3.81524078	6.238413e-01	sin(x)
100	-3.79005767	6.039637e-01	sin(x)
101	-3.76487456	5.837031e-01	sin(x)

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103  -3.71450835  5.420845e-01 sin(x)
104  -3.68932524  5.207529e-01 sin(x)
105  -3.66414213  4.990910e-01 sin(x)
106  -3.63895903  4.771127e-01 sin(x)
107  -3.61377592  4.548317e-01 sin(x)
108  -3.58859281  4.322624e-01 sin(x)
109  -3.56340970  4.094189e-01 sin(x)
110  -3.53822660  3.863158e-01 sin(x)
111  -3.51304349  3.629677e-01 sin(x)
112  -3.48786038  3.393894e-01 sin(x)
113  -3.46267727  3.155959e-01 sin(x)
114  -3.43749417  2.916023e-01 sin(x)
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116  -3.38712795  2.430756e-01 sin(x)
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118  -3.33676174  1.939324e-01 sin(x)
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121  -3.26121241  1.193347e-01 sin(x)
122  -3.23602931  9.429635e-02 sin(x)
123  -3.21084620  6.919820e-02 sin(x)
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125  -3.16047998  1.888621e-02 sin(x)
126  -3.13529688 -6.295735e-03 sin(x)
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128  -3.08493066 -5.663168e-02 sin(x)
129  -3.05974755 -8.175375e-02 sin(x)
130  -3.03456445 -1.068240e-01 sin(x)
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132  -2.98419823 -1.567454e-01 sin(x)
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137  -2.85828269 -2.795352e-01 sin(x)
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139  -2.80791648 -3.275186e-01 sin(x)
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167  -2.10278947 -8.617978e-01 sin(x)
168  -2.07760636 -8.742973e-01 sin(x)
169  -2.05242326 -8.862425e-01 sin(x)
170  -2.02724015 -8.976256e-01 sin(x)
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178 -1.82577529 -9.676686e-01 sin(x)
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180 -1.77540907 -9.791397e-01 sin(x)
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182 -1.72504286 -9.881276e-01 sin(x)
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204 -1.17101450 -9.211459e-01 sin(x)
205 -1.14583139 -9.110532e-01 sin(x)
206 -1.12064828 -9.003827e-01 sin(x)
207 -1.09546517 -8.891412e-01 sin(x)
208 -1.07028207 -8.773359e-01 sin(x)
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233 -0.44070438 -4.265766e-01 sin(x)
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240  -0.26442263 -2.613520e-01 sin(x)
241  -0.23923952 -2.369639e-01 sin(x)
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243  -0.18887331 -1.877524e-01 sin(x)
244  -0.16369020 -1.629602e-01 sin(x)
245  -0.13850709 -1.380647e-01 sin(x)
246  -0.11332398 -1.130816e-01 sin(x)
247  -0.08814088 -8.802680e-02 sin(x)
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249  -0.03777466 -3.776568e-02 sin(x)
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252   0.03777466  3.776568e-02 sin(x)
253   0.06295777  6.291619e-02 sin(x)
254   0.08814088  8.802680e-02 sin(x)
255   0.11332398  1.130816e-01 sin(x)
256   0.13850709  1.380647e-01 sin(x)
257   0.16369020  1.629602e-01 sin(x)
258   0.18887331  1.877524e-01 sin(x)
259   0.21405641  2.124255e-01 sin(x)
260   0.23923952  2.369639e-01 sin(x)
261   0.26442263  2.613520e-01 sin(x)
262   0.28960574  2.855744e-01 sin(x)
263   0.31478884  3.096157e-01 sin(x)
264   0.33997195  3.334606e-01 sin(x)
265   0.36515506  3.570941e-01 sin(x)
266   0.39033817  3.805012e-01 sin(x)
267   0.41552127  4.036669e-01 sin(x)
268   0.44070438  4.265766e-01 sin(x)
269   0.46588749  4.492159e-01 sin(x)
270   0.49107060  4.715702e-01 sin(x)
271   0.51625370  4.936255e-01 sin(x)
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273   0.56661992  5.367833e-01 sin(x)
274   0.59180302  5.578583e-01 sin(x)
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285   0.86881721  7.635657e-01 sin(x)
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291   1.01991585  8.520640e-01 sin(x)
292   1.04509896  8.649742e-01 sin(x)
293   1.07028207  8.773359e-01 sin(x)
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295   1.12064828  9.003827e-01 sin(x)
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301   1.27174693  9.556170e-01 sin(x)
302   1.29693003  9.627324e-01 sin(x)
303   1.32211314  9.692374e-01 sin(x)
304   1.34729625  9.751277e-01 sin(x)
305   1.37247936  9.803996e-01 sin(x)
306   1.39766246  9.850497e-01 sin(x)
307   1.42284557  9.890752e-01 sin(x)
308   1.44802868  9.924735e-01 sin(x)
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309   1.47321179  9.952424e-01 sin(x)
310   1.49839489  9.973802e-01 sin(x)
311   1.52357800  9.988854e-01 sin(x)
312   1.54876111  9.997572e-01 sin(x)
313   1.57394422  9.999950e-01 sin(x)
314   1.59912732  9.995987e-01 sin(x)
315   1.62431043  9.985685e-01 sin(x)
316   1.64949354  9.969050e-01 sin(x)
317   1.67467664  9.946093e-01 sin(x)
318   1.69985975  9.916829e-01 sin(x)
319   1.72504286  9.881276e-01 sin(x)
320   1.75022597  9.839456e-01 sin(x)
321   1.77540907  9.791397e-01 sin(x)
322   1.80059218  9.737129e-01 sin(x)
323   1.82577529  9.676686e-01 sin(x)
324   1.85095840  9.610106e-01 sin(x)
325   1.87614150  9.537432e-01 sin(x)
326   1.90132461  9.458710e-01 sin(x)
327   1.92650772  9.373990e-01 sin(x)
328   1.95169083  9.283325e-01 sin(x)
329   1.97687393  9.186773e-01 sin(x)
330   2.00205704  9.084395e-01 sin(x)
331   2.02724015  8.976256e-01 sin(x)
332   2.05242326  8.862425e-01 sin(x)
333   2.07760636  8.742973e-01 sin(x)
334   2.10278947  8.617978e-01 sin(x)
335   2.12797258  8.487517e-01 sin(x)
336   2.15315569  8.351673e-01 sin(x)
337   2.17833879  8.210534e-01 sin(x)
338   2.20352190  8.064188e-01 sin(x)
339   2.22870501  7.912727e-01 sin(x)
340   2.25388812  7.756249e-01 sin(x)
341   2.27907122  7.594852e-01 sin(x)
342   2.30425433  7.428639e-01 sin(x)
343   2.32943744  7.257715e-01 sin(x)
344   2.35462055  7.082189e-01 sin(x)
345   2.37980365  6.902171e-01 sin(x)
346   2.40498676  6.717776e-01 sin(x)
347   2.43016987  6.529121e-01 sin(x)
348   2.45535298  6.336326e-01 sin(x)
349   2.48053608  6.139512e-01 sin(x)
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351   2.53090230  5.734332e-01 sin(x)
352   2.55608541  5.526222e-01 sin(x)
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355   2.63163473  4.881405e-01 sin(x)
356   2.65681784  4.660091e-01 sin(x)
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358   2.70718405  4.208740e-01 sin(x)
359   2.73236716  3.978989e-01 sin(x)
360   2.75755027  3.746715e-01 sin(x)
361   2.78273337  3.512064e-01 sin(x)
362   2.80791648  3.275186e-01 sin(x)
363   2.83309959  3.036232e-01 sin(x)
364   2.85828269  2.795352e-01 sin(x)
365   2.88346580  2.552699e-01 sin(x)
366   2.90864891  2.308428e-01 sin(x)
367   2.93383202  2.062692e-01 sin(x)
368   2.95901512  1.815649e-01 sin(x)
369   2.98419823  1.567454e-01 sin(x)
370   3.00938134  1.318265e-01 sin(x)
371   3.03456445  1.068240e-01 sin(x)
372   3.05974755  8.175375e-02 sin(x)
373   3.08493066  5.663168e-02 sin(x)
374   3.11011377  3.147369e-02 sin(x)
375   3.13529688  6.295735e-03 sin(x)
376   3.16047998 -1.888621e-02 sin(x)
377   3.18566309 -4.405617e-02 sin(x)
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379   3.23602931 -9.429635e-02 sin(x)
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381   3.28639552 -1.442974e-01 sin(x)
382   3.31157863 -1.691685e-01 sin(x)
383   3.33676174 -1.939324e-01 sin(x)
384   3.36194484 -2.185733e-01 sin(x)
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386   3.41231106 -2.674237e-01 sin(x)
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706  -1.12064828  4.350988e-01 cos(x)
707  -1.09546517  4.576329e-01 cos(x)
708  -1.07028207  4.798768e-01 cos(x)
709  -1.04509896  5.018163e-01 cos(x)
710  -1.01991585  5.234377e-01 cos(x)
711  -0.99473274  5.447270e-01 cos(x)
712  -0.96954964  5.656710e-01 cos(x)
713  -0.94436653  5.862562e-01 cos(x)
714  -0.91918342  6.064696e-01 cos(x)
715  -0.89400031  6.262985e-01 cos(x)
716  -0.86881721  6.457301e-01 cos(x)
717  -0.84363410  6.647523e-01 cos(x)
718  -0.81845099  6.833529e-01 cos(x)
719  -0.79326788  7.015202e-01 cos(x)
720  -0.76808478  7.192426e-01 cos(x)
721  -0.74290167  7.365089e-01 cos(x)
722  -0.71771856  7.533081e-01 cos(x)
```

```
723  -0.69253545  7.696296e-01 cos(x)
724  -0.66735235  7.854631e-01 cos(x)
725  -0.64216924  8.007984e-01 cos(x)
726  -0.61698613  8.156259e-01 cos(x)
727  -0.59180302  8.299362e-01 cos(x)
728  -0.56661992  8.437202e-01 cos(x)
729  -0.54143681  8.569691e-01 cos(x)
730  -0.51625370  8.696745e-01 cos(x)
731  -0.49107060  8.818285e-01 cos(x)
732  -0.46588749  8.934232e-01 cos(x)
733  -0.44070438  9.044514e-01 cos(x)
734  -0.41552127  9.149060e-01 cos(x)
735  -0.39033817  9.247804e-01 cos(x)
736  -0.36515506  9.340684e-01 cos(x)
737  -0.33997195  9.427640e-01 cos(x)
738  -0.31478884  9.508618e-01 cos(x)
739  -0.28960574  9.583565e-01 cos(x)
740  -0.26442263  9.652436e-01 cos(x)
741  -0.23923952  9.715185e-01 cos(x)
742  -0.21405641  9.771773e-01 cos(x)
743  -0.18887331  9.822164e-01 cos(x)
744  -0.16369020  9.866326e-01 cos(x)
745  -0.13850709  9.904232e-01 cos(x)
746  -0.11332398  9.935857e-01 cos(x)
747  -0.08814088  9.961181e-01 cos(x)
748  -0.06295777  9.980188e-01 cos(x)
749  -0.03777466  9.992866e-01 cos(x)
750  -0.01259155  9.999207e-01 cos(x)
751   0.01259155  9.999207e-01 cos(x)
752   0.03777466  9.992866e-01 cos(x)
753   0.06295777  9.980188e-01 cos(x)
754   0.08814088  9.961181e-01 cos(x)
755   0.11332398  9.935857e-01 cos(x)
756   0.13850709  9.904232e-01 cos(x)
757   0.16369020  9.866326e-01 cos(x)
758   0.18887331  9.822164e-01 cos(x)
759   0.21405641  9.771773e-01 cos(x)
760   0.23923952  9.715185e-01 cos(x)
761   0.26442263  9.652436e-01 cos(x)
762   0.28960574  9.583565e-01 cos(x)
763   0.31478884  9.508618e-01 cos(x)
764   0.33997195  9.427640e-01 cos(x)
765   0.36515506  9.340684e-01 cos(x)
766   0.39033817  9.247804e-01 cos(x)
767   0.41552127  9.149060e-01 cos(x)
768   0.44070438  9.044514e-01 cos(x)
769   0.46588749  8.934232e-01 cos(x)
770   0.49107060  8.818285e-01 cos(x)
771   0.51625370  8.696745e-01 cos(x)
772   0.54143681  8.569691e-01 cos(x)
773   0.56661992  8.437202e-01 cos(x)
774   0.59180302  8.299362e-01 cos(x)
775   0.61698613  8.156259e-01 cos(x)
776   0.64216924  8.007984e-01 cos(x)
777   0.66735235  7.854631e-01 cos(x)
778   0.69253545  7.696296e-01 cos(x)
779   0.71771856  7.533081e-01 cos(x)
780   0.74290167  7.365089e-01 cos(x)
781   0.76808478  7.192426e-01 cos(x)
782   0.79326788  7.015202e-01 cos(x)
783   0.81845099  6.833529e-01 cos(x)
784   0.84363410  6.647523e-01 cos(x)
785   0.86881721  6.457301e-01 cos(x)
786   0.89400031  6.262985e-01 cos(x)
787   0.91918342  6.064696e-01 cos(x)
788   0.94436653  5.862562e-01 cos(x)
789   0.96954964  5.656710e-01 cos(x)
790   0.99473274  5.447270e-01 cos(x)
791   1.01991585  5.234377e-01 cos(x)
```

```
792   1.04509896  5.018163e-01 cos(x)
793   1.07028207  4.798768e-01 cos(x)
794   1.09546517  4.576329e-01 cos(x)
795   1.12064828  4.350988e-01 cos(x)
796   1.14583139  4.122888e-01 cos(x)
797   1.17101450  3.892174e-01 cos(x)
798   1.19619760  3.658991e-01 cos(x)
799   1.22138071  3.423488e-01 cos(x)
800   1.24656382  3.185814e-01 cos(x)
801   1.27174693  2.946119e-01 cos(x)
802   1.29693003  2.704557e-01 cos(x)
803   1.32211314  2.461279e-01 cos(x)
804   1.34729625  2.216440e-01 cos(x)
805   1.37247936  1.970196e-01 cos(x)
806   1.39766246  1.722702e-01 cos(x)
807   1.42284557  1.474116e-01 cos(x)
808   1.44802868  1.224595e-01 cos(x)
809   1.47321179  9.742974e-02 cos(x)
810   1.49839489  7.233820e-02 cos(x)
811   1.52357800  4.720078e-02 cos(x)
812   1.54876111  2.203344e-02 cos(x)
813   1.57394422 -3.147883e-03 cos(x)
814   1.59912732 -2.832721e-02 cos(x)
815   1.62431043 -5.348857e-02 cos(x)
816   1.64949354 -7.861600e-02 cos(x)
817   1.67467664 -1.036936e-01 cos(x)
818   1.69985975 -1.287054e-01 cos(x)
819   1.72504286 -1.536356e-01 cos(x)
820   1.75022597 -1.784684e-01 cos(x)
821   1.77540907 -2.031880e-01 cos(x)
822   1.80059218 -2.277788e-01 cos(x)
823   1.82577529 -2.522251e-01 cos(x)
824   1.85095840 -2.765114e-01 cos(x)
825   1.87614150 -3.006224e-01 cos(x)
826   1.90132461 -3.245428e-01 cos(x)
827   1.92650772 -3.482573e-01 cos(x)
828   1.95169083 -3.717510e-01 cos(x)
829   1.97687393 -3.950090e-01 cos(x)
830   2.00205704 -4.180164e-01 cos(x)
831   2.02724015 -4.407588e-01 cos(x)
832   2.05242326 -4.632216e-01 cos(x)
833   2.07760636 -4.853907e-01 cos(x)
834   2.10278947 -5.072520e-01 cos(x)
835   2.12797258 -5.287916e-01 cos(x)
836   2.15315569 -5.499959e-01 cos(x)
837   2.17833879 -5.708514e-01 cos(x)
838   2.20352190 -5.913449e-01 cos(x)
839   2.22870501 -6.114634e-01 cos(x)
840   2.25388812 -6.311941e-01 cos(x)
841   2.27907122 -6.505245e-01 cos(x)
842   2.30425433 -6.694425e-01 cos(x)
843   2.32943744 -6.879358e-01 cos(x)
844   2.35462055 -7.059930e-01 cos(x)
845   2.37980365 -7.236024e-01 cos(x)
846   2.40498676 -7.407529e-01 cos(x)
847   2.43016987 -7.574337e-01 cos(x)
848   2.45535298 -7.736341e-01 cos(x)
849   2.48053608 -7.893440e-01 cos(x)
850   2.50571919 -8.045533e-01 cos(x)
851   2.53090230 -8.192523e-01 cos(x)
852   2.55608541 -8.334319e-01 cos(x)
853   2.58126851 -8.470829e-01 cos(x)
854   2.60645162 -8.601967e-01 cos(x)
855   2.63163473 -8.727650e-01 cos(x)
856   2.65681784 -8.847799e-01 cos(x)
857   2.68200094 -8.962337e-01 cos(x)
858   2.70718405 -9.071191e-01 cos(x)
859   2.73236716 -9.174293e-01 cos(x)
860   2.75755027 -9.271576e-01 cos(x)
```

```
861   2.78273337 -9.362981e-01 cos(x)
862   2.80791648 -9.448447e-01 cos(x)
863   2.83309959 -9.527922e-01 cos(x)
864   2.85828269 -9.601354e-01 cos(x)
865   2.88346580 -9.668698e-01 cos(x)
866   2.90864891 -9.729911e-01 cos(x)
867   2.93383202 -9.784953e-01 cos(x)
868   2.95901512 -9.833790e-01 cos(x)
869   2.98419823 -9.876390e-01 cos(x)
870   3.00938134 -9.912728e-01 cos(x)
871   3.03456445 -9.942779e-01 cos(x)
872   3.05974755 -9.966526e-01 cos(x)
873   3.08493066 -9.983951e-01 cos(x)
874   3.11011377 -9.995046e-01 cos(x)
875   3.13529688 -9.999802e-01 cos(x)
876   3.16047998 -9.998216e-01 cos(x)
877   3.18566309 -9.990291e-01 cos(x)
878   3.21084620 -9.976029e-01 cos(x)
879   3.23602931 -9.955442e-01 cos(x)
880   3.26121241 -9.928541e-01 cos(x)
881   3.28639552 -9.895344e-01 cos(x)
882   3.31157863 -9.855871e-01 cos(x)
883   3.33676174 -9.810149e-01 cos(x)
884   3.36194484 -9.758205e-01 cos(x)
885   3.38712795 -9.700073e-01 cos(x)
886   3.41231106 -9.635790e-01 cos(x)
887   3.43749417 -9.565396e-01 cos(x)
888   3.46267727 -9.488937e-01 cos(x)
889   3.48786038 -9.406460e-01 cos(x)
890   3.51304349 -9.318017e-01 cos(x)
891   3.53822660 -9.223666e-01 cos(x)
892   3.56340970 -9.123465e-01 cos(x)
893   3.58859281 -9.017479e-01 cos(x)
894   3.61377592 -8.905774e-01 cos(x)
895   3.63895903 -8.788421e-01 cos(x)
896   3.66414213 -8.665496e-01 cos(x)
897   3.68932524 -8.537075e-01 cos(x)
898   3.71450835 -8.403240e-01 cos(x)
899   3.73969146 -8.264076e-01 cos(x)
900   3.76487456 -8.119672e-01 cos(x)
901   3.79005767 -7.970118e-01 cos(x)
902   3.81524078 -7.815510e-01 cos(x)
903   3.84042389 -7.655946e-01 cos(x)
904   3.86560699 -7.491527e-01 cos(x)
905   3.89079010 -7.322357e-01 cos(x)
906   3.91597321 -7.148543e-01 cos(x)
907   3.94115631 -6.970197e-01 cos(x)
908   3.96633942 -6.787430e-01 cos(x)
909   3.99152253 -6.600358e-01 cos(x)
910   4.01670564 -6.409101e-01 cos(x)
911   4.04188874 -6.213780e-01 cos(x)
912   4.06707185 -6.014518e-01 cos(x)
913   4.09225496 -5.811442e-01 cos(x)
914   4.11743807 -5.604681e-01 cos(x)
915   4.14262117 -5.394365e-01 cos(x)
916   4.16780428 -5.180629e-01 cos(x)
917   4.19298739 -4.963607e-01 cos(x)
918   4.21817050 -4.743438e-01 cos(x)
919   4.24335360 -4.520260e-01 cos(x)
920   4.26853671 -4.294216e-01 cos(x)
921   4.29371982 -4.065449e-01 cos(x)
922   4.31890293 -3.834104e-01 cos(x)
923   4.34408603 -3.600327e-01 cos(x)
924   4.36926914 -3.364267e-01 cos(x)
925   4.39445225 -3.126074e-01 cos(x)
926   4.41963536 -2.885898e-01 cos(x)
927   4.44481846 -2.643892e-01 cos(x)
928   4.47000157 -2.400209e-01 cos(x)
929   4.49518468 -2.155005e-01 cos(x)
```

```
930   4.52036779 -1.908433e-01 cos(x)
931   4.54555089 -1.660652e-01 cos(x)
932   4.57073400 -1.411817e-01 cos(x)
933   4.59591711 -1.162087e-01 cos(x)
934   4.62110022 -9.116202e-02 cos(x)
935   4.64628332 -6.605752e-02 cos(x)
936   4.67146643 -4.091113e-02 cos(x)
937   4.69664954 -1.573879e-02 cos(x)
938   4.72183265  9.443525e-03 cos(x)
939   4.74701575  3.461985e-02 cos(x)
940   4.77219886  5.977423e-02 cos(x)
941   4.79738197  8.489070e-02 cos(x)
942   4.82256508  1.099533e-01 cos(x)
943   4.84774818  1.349462e-01 cos(x)
944   4.87293129  1.598536e-01 cos(x)
945   4.89811440  1.846595e-01 cos(x)
946   4.92329751  2.093484e-01 cos(x)
947   4.94848061  2.339045e-01 cos(x)
948   4.97366372  2.583122e-01 cos(x)
949   4.99884683  2.825562e-01 cos(x)
950   5.02402993  3.066210e-01 cos(x)
951   5.04921304  3.304913e-01 cos(x)
952   5.07439615  3.541520e-01 cos(x)
953   5.09957926  3.775882e-01 cos(x)
954   5.12476236  4.007849e-01 cos(x)
955   5.14994547  4.237274e-01 cos(x)
956   5.17512858  4.464013e-01 cos(x)
957   5.20031169  4.687920e-01 cos(x)
958   5.22549479  4.908855e-01 cos(x)
959   5.25067790  5.126676e-01 cos(x)
960   5.27586101  5.341247e-01 cos(x)
961   5.30104412  5.552430e-01 cos(x)
962   5.32622722  5.760092e-01 cos(x)
963   5.35141033  5.964102e-01 cos(x)
964   5.37659344  6.164329e-01 cos(x)
965   5.40177655  6.360647e-01 cos(x)
966   5.42695965  6.552932e-01 cos(x)
967   5.45214276  6.741061e-01 cos(x)
968   5.47732587  6.924915e-01 cos(x)
969   5.50250898  7.104377e-01 cos(x)
970   5.52769208  7.279335e-01 cos(x)
971   5.55287519  7.449676e-01 cos(x)
972   5.57805830  7.615292e-01 cos(x)
973   5.60324141  7.776080e-01 cos(x)
974   5.62842451  7.931936e-01 cos(x)
975   5.65360762  8.082762e-01 cos(x)
976   5.67879073  8.228463e-01 cos(x)
977   5.70397384  8.368945e-01 cos(x)
978   5.72915694  8.504120e-01 cos(x)
979   5.75434005  8.633903e-01 cos(x)
980   5.77952316  8.758210e-01 cos(x)
981   5.80470627  8.876962e-01 cos(x)
982   5.82988937  8.990086e-01 cos(x)
983   5.85507248  9.097508e-01 cos(x)
984   5.88025559  9.199162e-01 cos(x)
985   5.90543870  9.294981e-01 cos(x)
986   5.93062180  9.384906e-01 cos(x)
987   5.95580491  9.468880e-01 cos(x)
988   5.98098802  9.546848e-01 cos(x)
989   6.00617113  9.618763e-01 cos(x)
990   6.03135423  9.684578e-01 cos(x)
991   6.05653734  9.744251e-01 cos(x)
992   6.08172045  9.797745e-01 cos(x)
993   6.10690356  9.845026e-01 cos(x)
994   6.13208666  9.886063e-01 cos(x)
995   6.15726977  9.920831e-01 cos(x)
996   6.18245288  9.949308e-01 cos(x)
997   6.20763598  9.971475e-01 cos(x)
998   6.23281909  9.987319e-01 cos(x)
```

```
999   6.25800220  9.996829e-01 cos(x)
1000  6.28318531  1.000000e+00 cos(x)
```

Step 3.1: Initialize the ggplot object

```
p <- ggplot(df, aes(x = x, y = y, color = group, linetype = group))
p
```



Step 3.2: Add the line Geometry

```
p <- p + geom_line(size = 1.2)
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead.
```

```
p
```



Step 3.3: Add plot Labels

```
p <- p + labs(title = "function curves : sin(x) and cos(x)",
              x = "x",
              y = "y = f(x)",
              color = "Function",
              linetype = "Function")
p
```



Step 3.4: Apply a Clean Theme

```
p <- p + theme_minimal()
p
```

